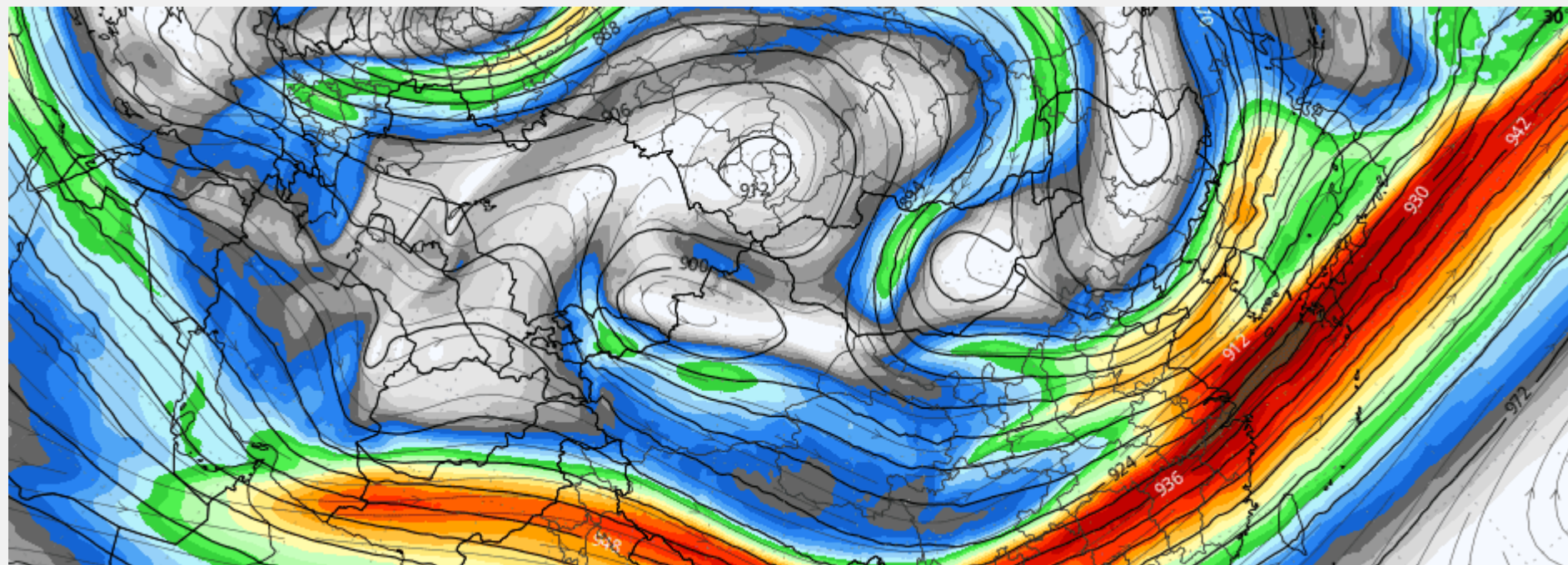


Climate Prediction Part 2

Project Objectives



- To find new patterns in weather data over the last 60 years
- To identify weather patterns outside the regional norm in Europe
- To determine whether unusual weather patterns are increasing
- To generate possibilities for future weather conditions over the next 25 to 50 years based on current trends
- To determine the safest places for people to live in Europe for the next 25 to 50 years



Summary:

If the sea level is expected to rise over the next 50 years, how will flooding patterns change as a result? Is it possible to use machine learning algorithms to predict areas most vulnerable to inundation, or even the expected water depths across flooding areas as they occur?

Best Architecture:

Convolutional Neural Networks trained on satellite imagery might identify coastal areas with features that make them especially vulnerable to flooding with higher sea levels and could make accurate predictions of the physical extent and potential damage from floods.

Data Requirements:

- Historical data on floods from past ~20 years.
- Satellite imagery from flooded areas over the same time period.



Summary:

If multi-variable weather extremes (such as simultaneously extremely high temperatures, low humidity, and high radiation) become more common, what effects could they have on water stress or crop yields? Can we predict the occurrence of these compound weather events and their impacts on society?

Best Architecture:

A Random Forest model can capture complex, nonlinear interactions, such as those that might lead to crop failure or drought. Furthermore, a Random Forest model can rank the importance of individual features that contribute to those conditions, allowing researchers and policymakers to better understand and maybe prevent potential shocks to food and water systems.

Data Requirements:

- Agricultural output measures and water stress indices from the same locations as our weather data.



Summary:

If natural disasters become more common and more extreme, to the point that certain highly-populated areas become uninhabitable, how can we expect refugee movement patterns to and within Europe to change?

Best Architecture:

A GAN is well-suited to this problem not only for their ability to uncover complex interactions between variables, but also because they generate novel data. This is crucial as the problem we're looking at is effectively unprecedented and so historical data can only be of limited usefulness. Additionally, a GAN can make predictions based on conditional statements, allowing for predictions tailored to expected conditions (e.g. "if Country X sees this much flooding, how many additional refugees can we expect in Country Y?").

Data Requirements:

- Historical data on refugee flows, especially international to Europe (e.g. the Syrian refugee crisis of 2011)
- Geopolitical features drawn from data including GDP, immigration policies, political violence and war, etc.
- Outputs from the previously-outlined machine learning models, which could indicate regions likely to have a large displaced populations

Final Recommendation

Random Forest Model for Crop Failure

While both of the first two thought experiments are practical and achievable with available resources, the second makes more direct use of ClimateWins' existing dataset. Furthermore, this application has potential to lead to important life-saving insights and strengthen Europe's resilience to one of the more devastating effects of global warming. The Random Forest algorithm's ability to capture nonlinear effects of compound weather events makes it ideally suited to this complex task and could provide researchers and other stakeholders with a much deeper understanding of the causes (and possible means of preventing) crop failure.

Questions?

You can check out more of my work on Tableau Public or GitHub at the links to the right.

Feel free to reach out if you'd like to know more!

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